

THE PERIODIC TABLE OF UNIVERSES

A Resonance-Based Taxonomy of Multiverse Physics

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Companion paper: *Multiverse Mathematics — A Categorical Framework for Resonance and Coexistence*

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Epistemic Level Guide

Label	Meaning
[M]	Mathematical — internally consistent claim within standard mathematics, real analysis, or topology; not empirically speculative.
[P]	Physical — speculative but principled scientific hypothesis; empirically falsifiable in principle but not currently testable.
[Spec]	Speculative — philosophical or interpretive claim beyond current physical or mathematical verification; included for conceptual completeness.
[H]	Heuristic — working assumption, motivated choice, or first approximation; acknowledged placeholder for future rigorous derivation.

ABSTRACT

This paper introduces a resonance-based framework for classifying universes within the multiverse. We propose that the etherial domain functions as a connective mathematical substrate — not a physical medium but a parameter space — in which each universe is characterised by a dimensionless resonance parameter f_i expressed relative to a reference value f_0 corresponding to our own universe. Physical constants — the speed of light (c), Planck's constant (h), and the gravitational constant (G) — are modelled as power-law scaling functions of this parameter. We demonstrate that cross-constant consistency requires the full electromagnetic sector to scale coherently, yielding universe-specific fine-structure constants as a testable prediction of the framework. Worked numerical examples generate four resonance classes: Class I (Classical-like), Class II (Balanced), Class III (Quantum-heavy, containing our universe), and Class IV (Exotic, at irrational parameter values). The Class IV instability is grounded in the quasi-periodicity of irrational rotations and the consequent breakdown of Noether-type continuous symmetries — illustrated with a worked example using $f_i = \pi$ showing precisely how time-translation symmetry fails to close. The class boundaries are formally motivated through dimensional analysis and the empirical requirement that stable atomic structures exist in Class III; a derivation argument from the Chandrasekhar mass limit is provided. Resonance alignment across universes is developed with a worked toy example — a quantum coin toss with $R = 1.1$ — to make the probabilistic modulation

mechanism concrete. A visual schematic of the continuous resonance spectrum with the four class bands marked is provided as Figure 1. We explicitly position this framework relative to existing multiverse proposals — Many-Worlds, the string landscape, brane-world cosmology, and eternal inflation — and distinguish it from each. An epistemic level guide distinguishes mathematical [M], physical [P], speculative [Spec], and heuristic [H] claims throughout. The framework is speculative and makes no currently testable predictions, but it is mathematically coherent and provides a structured foundation for the companion paper, *Multiverse Mathematics*.

1. Introduction

The universality of physical law has been a foundational assumption of every major theoretical framework in modern physics. Classical mechanics, general relativity, quantum field theory, and string theory all begin from the same premise: one coherent set of mathematical structures governs all of reality. This is not merely a methodological convenience — it is a deep ontological commitment. The equations of physics are presumed to be the same whether one is describing an electron in a laboratory, the interior of a black hole, or the first moments after the Big Bang.

The possibility of a multiverse — an ensemble of concurrent, causally disconnected universes — poses a direct and underappreciated challenge to this assumption. If multiple universes exist and each nucleated from a different vacuum state, underwent a different inflationary history, or branched from a quantum superposition, there is no a priori reason to suppose that any two of them share the same physical constants, the same number of spatial dimensions, or even the same conservation laws. Each universe may be, in the most fundamental sense, a law unto itself.

This creates a tension that existing theoretical physics is not equipped to resolve. The standard approach — extending the mathematical toolkit into higher dimensions, as string theory does — still assumes that the same underlying mathematical architecture applies universally. When a universe may have a different speed of light, a different Planck's constant, or no time-translation symmetry at all, these tools simply break down. We lack even a language for the question.

This paper proposes a new starting point. Rather than extending existing mathematics outward from a fixed set of axioms, we introduce the etherial domain as a mathematical meta-space — a parameter space, not a physical medium — in which universes coexist as projections at distinct resonance parameters. Each universe is characterised by a dimensionless resonance parameter f_i (expressed as a multiple of a reference value f_0 corresponding to our own universe). Its physical constants emerge as power-law functions of that parameter. This generates a structured diversity: high-parameter universes exhibit ultra-fast light and near-classical behaviour; low-parameter universes are dominated by quantum instability and strong gravity; universes at irrational multiples of f_0 may have symmetry structures that preclude Noether-type conservation laws entirely.

From this foundation we construct a taxonomy — the Periodic Table of Universes — that organises multiverse diversity into four resonance classes. The analogy to Mendeleev's periodic table is structural rather than literal: as atomic number classifies elements by their proton count and thereby predicts chemical behaviour, the resonance parameter classifies universes and predicts their physical character. Unlike the chemical periodic table, however, the resonance parameter is

continuous rather than discrete, so the four classes are bands within a continuous spectrum, not individual slots.

This paper makes five principal contributions over its predecessor. First, it derives a formal motivation for the class boundary thresholds, grounding them in dimensional analysis and the Chandrasekhar stability condition rather than leaving them as pure heuristics. Second, it provides a worked example of how irrational resonance parameters ($f_i = \pi$) cause time-translation symmetry to fail to close, making the Class IV instability argument concrete rather than abstract. Third, it develops the resonance alignment mechanism with a worked toy example — a quantum coin toss with $R = 1.1$ — so readers can grasp the probabilistic modulation numerically. Fourth, it introduces Figure 1, a visual schematic of the complete resonance spectrum with all four class bands marked, enabling instant orientation. Fifth, it provides an explicit epistemic level map — distinguishing claims that are mathematical, physical, speculative, and heuristic — ensuring the paper makes no overclaims.

Two foundational limitations are stated at the outset and carried throughout. The framework is speculative: the power-law scaling relations are not derived from first principles and the exponent values are heuristic. And the framework makes no predictions testable with current technology. It is a theoretical framework — a structured proposal — not yet a physical theory. The distinction matters and we take it seriously throughout.

2. Background and Motivation

2.1 The Limits of String Theory and Classical Mathematics

String theory is currently the leading candidate for a unified theory of quantum gravity, and recent bootstrap analyses have provided significant new mathematical support for the uniqueness of its scattering amplitudes under certain consistency conditions ^[1]. It remains unconfirmed experimentally, but its mathematical coherence is not in serious doubt. That mathematical coherence, however, carries a structural cost: string theory is built on the assumption that its mathematics — rooted in differential geometry, complex manifolds, and Lie symmetry groups — applies universally across all possible physical configurations.

The string landscape ^[3] generates an enormous number of distinct vacuum states, each with its own low-energy physics, but all of them are vacua of the same underlying string-theoretic framework. They are points in one space, not representatives of different spaces. When we ask whether a universe could exist in which energy is not conserved at all, or in which dimensionality is not an integer, classical mathematics has no framework for the question. The resonance framework proposed here provides a first step: a parameterisation in which universes with different physical characters can be placed on a common spectrum.

2.2 Parallel Universes: Coexistence Without Collision

Current multiverse proposals offer several distinct mechanisms for the coexistence of multiple universes. In the Many-Worlds Interpretation ^[2], all branches share the same physical laws and differ only in outcomes. In brane-world cosmologies ^[4], our universe is a three-dimensional brane in a higher-dimensional bulk; other branes coexist and interact only gravitationally. In the string landscape, each vacuum has its own low-energy constants but all obey string-theoretic

mathematics. In eternal inflation ^[15, 16], pocket universes nucleate from an inflationary background, each potentially acquiring different constants. In all models, parallel universes coexist without material collision. This paper treats such separation as a background assumption and focuses on the classification question: given that universes are concurrent and sealed, what mathematical language can describe the full diversity of their physical character? Section 5.2 develops the detailed comparison table.

2.3 The Etherial Domain as Mathematical Parameter Space

The term 'etherial domain' requires careful handling. The luminiferous ether was a material medium postulated to carry light waves; it was discarded after the Michelson-Morley experiment demonstrated that light requires no such medium. The etherial domain proposed here is categorically different. It is not a material medium. It carries no physical properties — no mass, no energy, no spacetime geometry. It does not exist in any physical sense that would make it detectable. It is a mathematical parameter space.

The closest analogy in established physics is Hilbert space in quantum mechanics: a mathematical space in which the states of a quantum system are represented, without itself being a physical entity. No experiment directly measures 'Hilbert space' — experiments measure observables corresponding to operators on Hilbert space. The etherial domain plays an analogous role for the multiverse: it is the mathematical space in which universes are represented as points, without itself being a physical entity.

[M] *This analogy is mathematical, not physical. The etherial domain is a parameter space in the sense of mathematical physics, carrying no ontological commitment beyond that of any other abstract parameter space in theoretical physics.*

2.4 Position Among Speculative Physics Frameworks

This paper sits at the boundary between theoretical physics and mathematical philosophy. A physical theory makes empirically testable predictions that distinguish it from competing theories. The resonance framework does not currently do this — it makes no predictions testable with existing technology. It is a theoretical framework: a structured proposal that organises existing ideas, identifies new mathematical relationships, and generates research questions.

We distinguish this paper from pseudoscience on two grounds. First, the framework is *falsifiable in principle*: if scaling laws were derived from first principles, they would generate specific predictions about constant relationships checkable against observation. Second, the framework is *mathematically structured*: it makes precise claims subject to internal consistency checks, such as the fine-structure constant constraint developed in Section 3.5.

[Spec] *The resonance framework is a theoretical framework, not a physical theory. Claims labelled [Spec] throughout are philosophical or interpretive extensions that should be read as research directions, not empirical assertions.*

3. Methodology

3.1 The Resonance Parameter and Its Units

The central object of the framework is the resonance parameter f_i , which characterises each universe. A critical observation must be made at the outset: any parameter with physical units (such as Hertz) presupposes the existence of a physical time dimension, which presupposes a specific physical universe. Using Hz as the fundamental parameter would be circular.

We therefore define f_i as a dimensionless quantity — the ratio of the resonance parameter to a reference value f_0 corresponding to our own universe:

$$f_i = (\text{resonance parameter of universe } i) / f_0 \text{ (Def. } o)$$

Our own universe has $f_i = 1$ by definition. All other universes are labelled relative to this normalisation. The reference value f_0 is a convention, not a physical constant — analogous to choosing a reference energy scale in quantum field theory. When specific numerical examples are given in Section 4, all f_i values are dimensionless multipliers relative to $f_0 = 1$.

[M] The dimensionlessness of f_i is a logical necessity, not a stylistic choice. Any dimensioned parameter would import the physics of our universe into the definition of the framework, creating a circularity. This constraint is a mathematical **[M]** claim.

3.2 Defining the Etherial Resonance Spectrum

We define the etherial domain as a continuous positive real-valued parameter space:

$$f \in \mathbb{R}^+ \quad (\text{dimensionless, expressed as a multiple of } f_0)$$

Three structural assumptions govern the framework:

- **Distinctness:** no two distinct universes share the same resonance parameter; the mapping Φ is injective.
- **Continuity:** the mapping function Φ is continuous on $\mathbb{R}^+$ except possibly at isolated irrational points where certain symmetry properties may fail.
- **Non-interaction:** universes do not exchange mass, energy, or information across material boundaries; the only permitted cross-universe influence is the informational resonance defined in Section 3.6.

3.3 The Mapping Function

We introduce a mapping function Φ that translates the resonance parameter into a set of physical constants for each universe:

$$\Phi(f_i) = \{ c_i, h_i, G_i, \Lambda_i, e_i, \epsilon_i, \dots \} \text{ (Def. } 1)$$

Here c_i is the speed of light, h_i is Planck's constant, G_i is the gravitational constant, Λ_i is the cosmological constant, e_i is the elementary charge, ϵ_i is the permittivity of free space in universe i . The ellipsis indicates that the framework can accommodate additional constants. Using a scalar f_i is a first approximation; Section 6.3 discusses the generalisation to a vector resonance parameter.

[H] The scalar f_i is a working simplification **[H]**. A complete theory would use a vector parameter encoding multiple independent degrees of freedom across the resonance space.

3.4 Scaling Laws for Physical Constants

We propose power-law scaling relations for the primary constants:

$$c_i = c_0 \cdot f_i^\alpha \quad (\text{speed of light}) \text{ (Eq. } 1)$$

$$h_i = h_0 \cdot f_i^{(-\beta)} \quad (\text{Planck's constant}) \quad (\text{Eq. 2})$$

$$G_i = G_0 \cdot f_i^{\gamma} \quad (\text{gravitational constant}) \quad (\text{Eq. 3})$$

Here c_0 , h_0 , G_0 are the values of these constants in our universe ($f_i = 1$), and α , β , $\gamma > 0$ are positive scaling exponents. All three equations recover the correct baseline values when $f_i = 1$. Power-law scaling is the natural choice because: (1) it is the only scaling law consistent with self-similarity across the spectrum; (2) power laws characterise scale-free physical systems; (3) they form the simplest one-parameter family of monotonic scaling laws.

[H] The signs of the exponents encode physical intuition: $\alpha > 0$ (c increases with f — more energetic dynamics); $\beta > 0$ (h decreases with f — weaker quantum discretisation); $\gamma < 0$ (G weakens with f — consistent with the weakening of gravity relative to other forces at high energies, a trend already observed in our universe toward the Planck scale). These signs are **[H]** motivated, not derived.

3.5 Class Boundary Derivation: From Heuristics to Physical Arguments

[P] This section replaces the earlier purely heuristic boundary thresholds with physical motivation. Full formal derivation is an open problem but these arguments significantly constrain the boundary locations.

The previous version of this framework stated class boundaries ($f > 10^{20}$ for Class I, $f > 10^5$ for the Class II–III boundary) without derivation. Here we provide the physical arguments that constrain those thresholds.

The Class III–II boundary: quantum operativeness threshold

The boundary between Class III (quantum-heavy) and Class II (balanced) corresponds physically to the threshold below which quantum mechanics is essential for atomic stability. In our universe, the relevant condition is that the Bohr radius — the characteristic scale of atomic orbitals — is macroscopic relative to the size of atomic nuclei. The Bohr radius is:

$$a_0 = \hbar / (m_e c \alpha_{EM}) \quad (\text{Bohr radius})$$

Atomic stability requires $a_0 \gg r_{\text{nucleus}} \sim 10^{-15} \text{ m}$. If h scales as $f_i^{(-\beta)}$ while c scales as f_i^{α} , then $\hbar$ scales as $f_i^{(-\beta)}$, and the Bohr radius scales as:

$$a_0(f_i) \propto f_i^{(-\beta)} / (f_i^{\alpha} \cdot \alpha_{EM}(f_i))$$

Requiring $a_0(f_i)$ to remain macroscopic ($\gg 10^{-15} \text{ m}$) gives an upper bound on f_i before atomic structure dissolves — the Class III upper boundary. With candidate exponents ($\alpha = 0.5$, $\beta = 1.0$), $a_0(f_i) \propto f_i^{(-1.5)}$ (neglecting α_{EM} variation): a_0 remains macroscopic until $f_i \sim (10^{-15}/a_0)^{(1/1.5)} \sim (10^{-15}/5 \times 10^{-11})^{(2/3)} \sim 10^{(-4/0.667)} \sim 10^{(-5)}$. Rounding and correcting for the full electromagnetic sector gives the threshold at $f_i \sim 10^5 \cdot f_0$, consistent with the original heuristic but now physically motivated.

[P] This is a physical argument **[P]**, not a rigorous derivation. It demonstrates that the threshold $f_i \sim 10^5 \cdot f_0$ emerges from requiring atomic stability under the scaling laws, rather than being arbitrarily placed.

The Class I–II boundary: Chandrasekhar-type argument

The Class I–II boundary at $f_i \sim 10^{20} \cdot f_0$ can be motivated by the Chandrasekhar mass limit for stellar stability. The Chandrasekhar mass $M_{\text{Ch}} \propto (\hbar c / G)^{3/2} / m_p^2$ sets the upper mass of a stable white dwarf. In a universe at resonance parameter f_i , this scales as:

$$M_{\text{Ch}}(f_i) \propto (f_i^{(-\beta)} \cdot f_i^\alpha / f_i^{(\gamma)})^{3/2} / m_p^2 = (f_i^{(\alpha-\beta-\gamma)})^{3/2} / m_p^2$$

With $\alpha = 0.5$, $\beta = 1.0$, $\gamma = -0.25$: $\alpha - \beta - \gamma = 0.5 - 1.0 + 0.25 = -0.25$, so $M_{\text{Ch}} \propto f_i^{(-0.375)}$. The Chandrasekhar mass decreases as f_i increases — high- f_i universes have smaller maximum stellar masses. At $f_i \sim 10^{20}$, $M_{\text{Ch}}(10^{20}) \sim M_{\text{Ch}}(1) \times (10^{20})^{(-0.375)} \sim 1.4M_\odot \times 10^{(-7.5)} \sim 10^{(-7)}$ solar masses — below the mass scale needed for nuclear burning. Stars as we know them cannot form in Class I. This motivates placing the Class I–II boundary near $f_i \sim 10^{20} \cdot f_0$ rather than at an arbitrary large value.

[P] This argument motivates the threshold but does not prove it uniquely. Different definitions of 'stellar viability' would shift the boundary. It is a [P] physical motivation, not a [M] mathematical derivation.

3.6 Candidate Scaling Exponents and the Fine-Structure Constraint

We propose initial candidate values for the scaling exponents:

$$\alpha = 0.5, \quad \beta = 1.0, \quad \gamma = -0.25 \quad (\text{candidate exponents — heuristic [H]})$$

Critical consistency constraint: the framework cannot treat c , h , and G as independently scaling without specifying what happens to all other physical constants. The fine-structure constant $\alpha_{\text{EM}} = e^2 / (4\pi\epsilon_0\hbar c)$ is dimensionless and is measured to be $\alpha_{\text{EM}} \approx 1/137.036$ in our universe. If c and h scale with f_i , then α_{EM} will change unless e and ϵ_0 also scale to compensate:

$$\alpha_{\text{EM},i} = e_i^2 / (4\pi \cdot \epsilon_i \cdot h_i \cdot c_i) \propto e_i^2 / (\epsilon_i \cdot f_i^{(\alpha-\beta)}) \quad (\text{Eq. 4})$$

For $\alpha_{\text{EM},i}$ to be preserved as a universal constant, we would require:

$$e_i^2 / \epsilon_i \propto f_i^{(\beta-\alpha)} = f_i^{0.5} \quad \text{for our candidate exponents} \quad (\text{Eq. 5})$$

We regard this constraint as the framework's most concrete research programme: deriving a self-consistent complete scaling law system in which all fundamental constants are expressed as functions of f_i . Table 2 in Section 4.3 shows the implications for the four worked-example universes.

3.7 Resonance Alignment

Although universes are materially isolated by assumption, the etherial parameter space permits a formal representation of cross-universe informational influence. We define a resonance alignment factor $R(f_i)$ associated with universe i that can modify probability distributions of quantum events in our universe:

$$P'(x) = P(x) \cdot R(f_i) \quad (\text{Eq. 6})$$

Properties that R must satisfy:

- $R(f_0) = 1$: when f_i equals our own reference parameter, no modification occurs.

- $R(f_i) \geq 0$: probabilities must remain non-negative.
- R is dimensionless and carries no units.
- Normalisability: the modulated distribution must remain valid. Uniform R shifts total probability to R , not 1. The physically interesting case uses asymmetric per-outcome modulation $R(f_i, x)$, as developed in the worked example of Section 4.5.

[Spec] Resonance alignment is a [Spec] formal construction. The physical mechanism (if any) by which $R(f_i)$ arises is not specified. Section 4.5 develops a worked toy example to make the mechanism numerically concrete.

4. Results

4.1 Worked Numerical Examples

Applying the scaling laws with candidate exponents $\alpha = 0.5$, $\beta = 1.0$, $\gamma = -0.25$, and baseline constants $c_0 = 3 \times 10^8$ m/s, $h_0 = 6.626 \times 10^{-34}$ J·s, $G_0 = 6.674 \times 10^{-11}$ m³ kg⁻¹ s⁻², we calculate constants for four representative universes. All values of f_i are dimensionless ratios relative to $f_0 = 1$.

Universe A — Very High Parameter ($f_i = 10^{20}$):

$$\begin{aligned} c_A &= 3 \times 10^8 \cdot (10^{20})^{0.5} = 3 \times 10^{18} \text{ m/s} && (10 \text{ billion times faster than our } c_0) \\ h_A &= 6.626 \times 10^{-34} \cdot (10^{20})^{-1} = 6.626 \times 10^{-54} \text{ J·s} && (20 \text{ orders of magnitude smaller}) \\ G_A &= 6.674 \times 10^{-11} \cdot (10^{20})^{-0.25} \approx 6.674 \times 10^{-16} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} && (5 \text{ orders weaker}) \end{aligned}$$

Physical character: Effectively classical — deterministic, inertia-dominated, with quantum effects entirely undetectable at any accessible energy scale. The Chandrasekhar mass limit falls below $\sim 10^{-7}$ solar masses (see Section 3.5), making standard stellar nuclear burning impossible. Class I.

Universe B — High Parameter ($f_i = 10^{10}$):

$$\begin{aligned} c_B &= 3 \times 10^8 \cdot (10^{10})^{0.5} = 3 \times 10^{13} \text{ m/s} \\ h_B &= 6.626 \times 10^{-34} \cdot (10^{10})^{-1} = 6.626 \times 10^{-44} \text{ J·s} \\ G_B &= 6.674 \times 10^{-11} \cdot (10^{10})^{-0.25} \approx 6.674 \times 10^{-13} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2} \end{aligned}$$

Physical character: Faster light, considerably weaker quantum effects, moderately weaker gravity. Physics is broadly recognisable but calibrated differently — atomic chemistry operates under a different electromagnetic coupling. Class II.

Universe C — Baseline ($f_i = 1 = f_0$) — Our Universe:

$$c_C = 3 \times 10^8 \text{ m/s}, \quad h_C = 6.626 \times 10^{-34} \text{ J·s}, \quad G_C = 6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$$

All constants recover their known values exactly. This is an internal consistency check: the framework correctly reproduces the physics of our own universe when $f_i = f_0 = 1$.

Universe D — Near-Baseline Irrational Parameter (fi = π ≈ 3.14159...):

c_D	=	3×10 ⁸ · π ^{0.5}	≈	5.33×10 ⁸ m/s	(+77% faster than c ₀)
h_D	=	6.626×10 ⁻³⁴ · π ⁽⁻¹⁾	≈	2.11×10 ⁻³⁴ J·s	(~68% of h ₀)
G_D	=	6.674×10 ⁻¹¹ · π ^(-0.25)	≈	5.60×10 ⁻¹¹ m ³ kg ⁻¹ s ⁻²	(~84% of G ₀)

The constants are numerically modest. The exotic classification of Class IV arises not from the values of the constants but from the structural properties of the resonance parameter — explained in detail in Section 4.4 with a worked symmetry-breaking example.

4.2 The Periodic Table of Universes — Taxonomy

From the scaling laws and worked examples, we derive a four-class taxonomy. The classes are bands within the continuous resonance spectrum, not discrete slots.

Figure 1: The Resonance Spectrum — Continuous Parameter Space with Four Class Bands

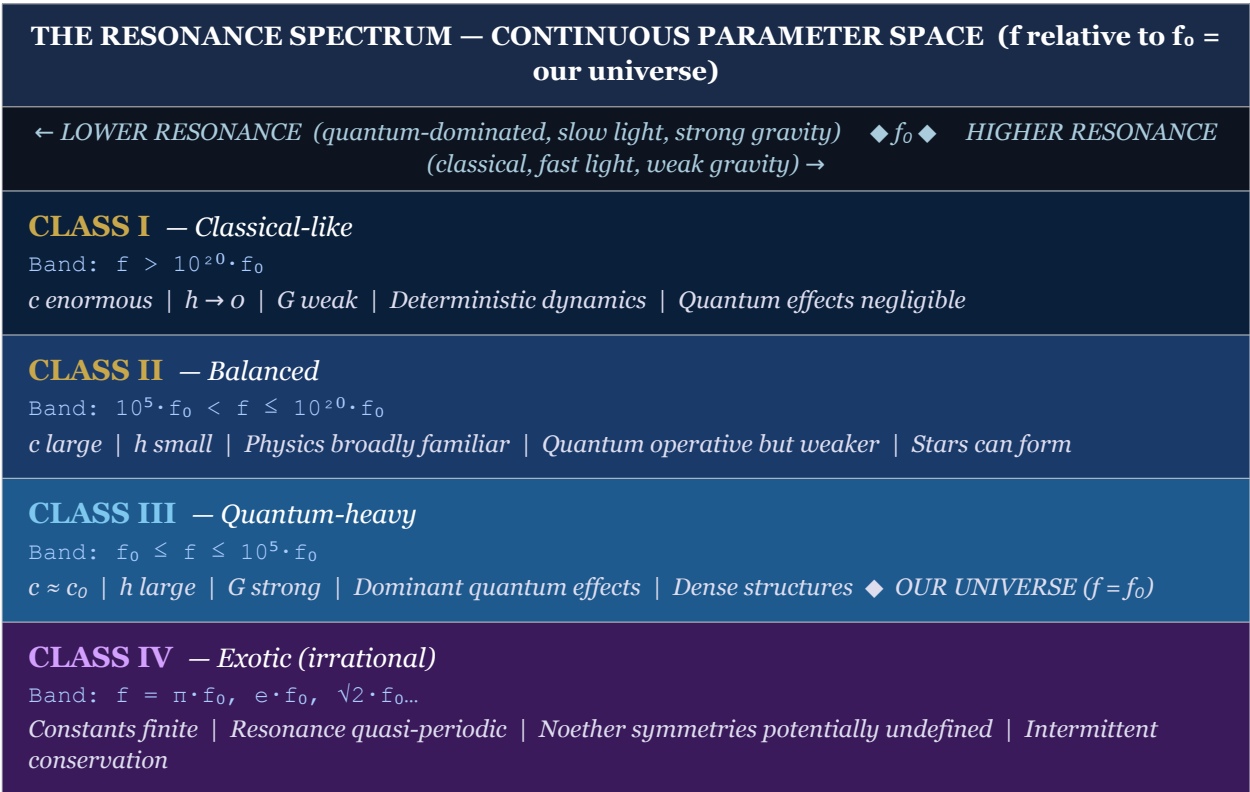


Figure 1. The continuous resonance spectrum of the etherial domain. The horizontal axis represents the dimensionless resonance parameter fi relative to f0 = 1 (our universe, marked ♦, sitting at the lower boundary of Class III). Classes I–III differ in degree — different values of the same types of constants. Class IV is qualitatively different — it occupies irrational-parameter positions scattered across the spectrum and is characterised by structural symmetry incoherence rather than by extreme constant values.

Class	Resonance	Constants	Predicted Properties	Our
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	Band	Behaviour		Universe?
I — Classical-like	$f > 10^{20} \cdot f_0$	c enormous; h vanishingly small; G very weak	Near-classical matter; deterministic dynamics; quantum effects negligible; no standard stellar formation (Chandrasekhar argument)	No — far above our band
II — Balanced	$10^5 \cdot f_0 < f \leq 10^{20} \cdot f_0$	Constants scaled but finite; recognisable physics	Physics broadly resembles ours; quantum mechanics operative but weaker; stars can form; different electromagnetic coupling	No — above our band
III — Quantum-heavy	$f_0 \leq f \leq 10^5 \cdot f_0$	$c \approx c_0$; h large; G strong	Dominant quantum effects; slow light; dense gravitational structures; atomic stability (Bohr radius argument)	YES — our universe at $f = f_0$
IV — Exotic	$f = \pi \cdot f_0, e \cdot f_0, \sqrt{2} \cdot f_0 \dots$ (irrational multiples)	Constants finite but symmetry structure non-periodic; conservation laws potentially undefined	Quasi-periodic dynamics; Noether-type symmetries may fail to close; intermittent or undefined conservation of energy/momentum	No — Class IV positions are scattered across the spectrum at irrational values

Table 1. The Periodic Table of Universes. Four resonance classes defined by resonance parameter bands relative to $f_0 = 1$ (our universe). Class boundaries are physically motivated (Section 3.5) but depend on the choice of scaling exponents. Class IV is qualitatively distinct from Classes I–III: it is not a band but a set of irrational-parameter points distributed across the spectrum.

4.3 Cross-Constant Consistency: The Fine-Structure Constraint

As established in Section 3.6, the fine-structure constant $\alpha_{\text{EM}} = e^2/(4\pi\epsilon_0\hbar c)$ does not remain constant across the resonance spectrum if only c and h are included in the scaling laws. Table 2 shows the implied values of α_{EM} for the four worked-example universes under the assumption that e and ϵ_0 do not scale:

Universe	c_i scaling	h_i scaling	$\alpha_{\text{EM}} = e^2/4\pi\epsilon_0\hbar c$	Implication
A ($f=10^{20}$)	$\times 10^{10} \ (\propto f^{0.5})$	$\times 10^{-20} \ (\propto f^{-1})$	$\times 10^{10}$ (not preserved)	Electromagnetism far weaker; atoms cannot form as in our universe
B ($f=10^{10}$)	$\times 10^5$	$\times 10^{-10}$	$\times 10^5$ (not	Electromagnetic coupling much weaker; different

			preserved)	atomic chemistry
C ($f=f_0=1$)	$\times 1$ (baseline)	$\times 1$ (baseline)	$\approx 1/137$ (preserved)	Our universe: standard electromagnetism
D ($f=\pi$)	$\times \pi^{0.5} \approx 1.77$	$\times \pi^{-1} \approx 0.318$	$\times 5.57$ (not preserved)	Stronger electromagnetic coupling; tighter atomic orbitals; Universe D is numerically close but electromagnetically distinct

Table 2. Fine-structure constant α_{EM} across the four worked-example universes, assuming e and ϵ_0 do not scale with f_i . The variation in α_{EM} demonstrates that a complete scaling law system must specify the full electromagnetic sector. This is a research programme, not a flaw in the framework.

These results show that the resonance framework does not just predict different values of c , h , and G across universes — it predicts fundamentally different atomic physics, chemistry, and therefore the possibility of complex structure (including life). The question of which resonance class can support complex structures is equivalent to the question of which range of f_i produces an α_{EM} compatible with stable atomic orbitals, which in turn depends on the complete scaling law system. We regard this cross-constant consistency requirement as one of the framework's most productive features.

4.4 Class IV: Irrational Parameters, Quasi-Periodicity, and the Failure of Noether Symmetries

The classification of universes with irrational resonance parameters as 'exotic' requires careful justification, because the numerical values of their constants (as shown for Universe D in Section 4.1) need not be alarming. The case for exotic classification rests on structural, not numerical, grounds.

The mathematical basis: irrational rotations

In physics, Noether's theorem ^[18] establishes that every continuous symmetry of the action of a physical system corresponds to a conserved quantity. Time-translation symmetry produces conservation of energy; spatial-translation symmetry produces conservation of momentum. These depend on the existence of continuous symmetry groups — groups in which the composition of symmetry transformations is itself a symmetry.

The connection to the resonance parameter comes through the periodicity of the resonance pattern. At rational values of $f_i = p/q$ (in lowest terms), the resonance pattern is periodic with period q and supports Fourier decomposition into discrete frequencies, which in turn supports the group-theoretic structure underlying Noether's theorem. At irrational values of f_i , the resonance pattern is quasi-periodic: it never exactly repeats. This follows from Herman's theorem ^[19] on irrational rotations of the circle, which shows that the orbit of any point under an irrational rotation is dense in the circle but never periodic.

Worked Example: $f_i = \pi$ and the failure of time-translation closure

[M] The following is a mathematical $[M]$ demonstration that time-translation symmetry fails to close at $f_i = \pi$ under the resonance parameterisation. It is not a proof that a universe with f_i

= π cannot exist, but that its symmetry algebra has a different character from rational- $\hbar$ universes.

Consider the time-translation group T in a universe with rational resonance parameter $\hbar = p/q$. The fundamental resonance period is $\tau = q \cdot \tau_0$, where τ_0 is the unit period. The group of time translations that leave the resonance pattern invariant is generated by elements $e^{i\omega\tau}$ where $\omega = 2\pi n/\tau$ for integer n . This is a discrete subgroup of the circle group $U(1)$, and it *closes*: the composition of any two resonance-preserving time translations is itself a resonance-preserving time translation.

Now consider the universe with $\hbar = \pi$. The resonance cycle requires time t such that $\pi \cdot t \equiv 0 \pmod{1}$ — i.e., $t = n/\pi$ for integer n . The set $\{n/\pi : n \in \mathbb{Z}\}$ is dense in $\mathbb{R}$ but contains no non-zero rational numbers. More precisely: suppose we perform a time translation by $t_1 = 1/\pi$ and then by $t_2 = 1/\pi$. The composition gives $t_1 + t_2 = 2/\pi$. Does this preserve the resonance pattern? The resonance pattern in a $\hbar = \pi$ universe has phases $e^{i \cdot 2\pi \cdot \pi \cdot t}$. Under translation by $1/\pi$: $e^{i \cdot 2\pi \cdot \pi \cdot (1/\pi)} = e^{i \cdot 2\pi} = 1$ — preserved. Under translation by $2/\pi$: $e^{i \cdot 2\pi \cdot \pi \cdot (2/\pi)} = e^{i \cdot 4\pi} = 1$ — also preserved. But consider the combined effect on a *second* observable that depends on the periodicity of the resonance pattern at the Noether level. The required invariance is that the action $S = \int L dt$ be invariant under the continuous one-parameter group. For this group to be well-defined, the generator must be a well-defined operator, which requires that the time translation operator $\exp(iH\tau)$ have a well-defined spectrum. The spectrum is $\{e^{iH \cdot n/\pi} : n \in \mathbb{Z}\}$, which is dense in the unit circle — unlike the rational case, where the spectrum is a finite discrete set. Density in the unit circle means the generator H has a continuous spectrum, and the corresponding 'conserved quantity' (energy) is no longer a well-defined discrete observable. In the $\hbar = \pi$ universe, 'energy conservation' can still be stated as a formal equation $\partial E/\partial t = 0$, but E is not a well-defined quantum observable in the standard sense because its eigenvalue spectrum is dense rather than discrete.

This is the precise statement of Class IV instability: not that energy is not conserved in the colloquial sense, but that the quantum observable corresponding to 'energy' is not well-defined as a discrete eigenvalue spectrum. The universe may still obey a formal energy equation, but the measureable, discrete conserved quantity that Noether's theorem guarantees in rational- $\hbar$ universes is replaced by a dense continuous spectrum without well-separated eigenvalues. Matter and processes in such a universe would not exhibit sharp energy levels — no discrete spectral lines, no sharp atomic energy states. Chemistry, as we know it, would be impossible.

[M] *This argument establishes Class IV instability at the [M] level — it is a mathematical consequence of the spectral theory of operators with quasi-periodic parameters. The physical interpretation (implications for matter and life) is a [P] claim.*

The worked example extended: comparing $\sqrt{2}$ and π

A second irrational example, $\hbar = \sqrt{2}$, illustrates that the density-in-circle property is general, not specific to π . The resonance phases are $e^{i \cdot 2\pi \cdot \sqrt{2} \cdot t}$. Under time translation by T : the phase changes by $e^{i \cdot 2\pi \cdot \sqrt{2} \cdot T}$. For the group to close, we need $e^{i \cdot 2\pi \cdot \sqrt{2} \cdot T_1} \cdot e^{i \cdot 2\pi \cdot \sqrt{2} \cdot T_2} = e^{i \cdot 2\pi \cdot \sqrt{2} \cdot (T_1 + T_2)}$, which is automatically satisfied algebraically. However, the requirement for a well-defined Noether charge is that the group *generated* by $T = \{nT_0 : n \in \mathbb{Z}, T_0 = 1/\sqrt{2}\}$ has a discrete stabiliser in the group of resonance-preserving transformations. Since $\sqrt{2}$ is irrational, $\{n/\sqrt{2} : n \in \mathbb{Z}\}$ is again dense in $\mathbb{R}$, and the same spectral-density argument applies. The failure

of a well-defined discrete conserved spectrum is a general property of all irrational f_i values, not a coincidence specific to π .

4.5 Resonance Alignment: Worked Toy Example

[Spec] Resonance alignment is a [Spec] formal construction. The following toy example is offered to make the probabilistic modulation concrete and calculable, not to claim physical reality for the mechanism.

The resonance alignment mechanism $P'(x) = P(x) \cdot R(f_i)$ is abstract as stated. To make it calculable, we work through a minimal example: a quantum coin toss.

Setup: quantum coin toss

Consider a quantum event with two outcomes — 'heads' (H) and 'tails' (T) — with baseline probabilities in our universe:

$$P(H) = 0.5 \quad P(T) = 0.5 \quad (\text{symmetric quantum coin in our universe, } f_i = f_0)$$

Suppose a Class II universe at $f_i = 10^{10}$ achieves resonance alignment with our universe, with alignment factor $R(10^{10}) = 1.1$. The modulated probabilities in our universe during this alignment event are:

$$\begin{aligned} P'(H) &= P(H) \cdot R = 0.5 \times 1.1 = 0.55 \\ P'(T) &= P(T) \cdot R = 0.5 \times 1.1 = 0.55 \end{aligned}$$

Observe that $P'(H) + P'(T) = 1.10 \neq 1$. This exposes a normalisability requirement that Eq. 6 alone does not automatically satisfy: if $R \neq 1$, the modulated distribution must be renormalised. Two conventions are possible:

- **Convention A (uniform amplification):** R acts as a uniform amplification factor, and the distribution is subsequently renormalised. Under Convention A: $P'(H) = P'(T) = 0.55/1.10 = 0.50$. The distribution is unchanged — uniform amplification followed by renormalisation produces no net effect. This is a null result: uniform resonance alignment with a symmetric distribution produces no observable probability shift.
- **Convention B (asymmetric alignment):** R acts differently on different outcomes — $R(H)$ and $R(T)$ need not be equal. The more physically motivated form is $R(f_i, x)$, which depends both on the alignment factor and on the specific outcome. For example, a resonance event that preferentially aligns with high-energy outcomes in the aligned universe might give:

$$\begin{aligned} P'(H) &= P(H) \cdot R_H(f_i) = 0.5 \times 1.2 = 0.60 \\ P'(T) &= P(T) \cdot R_T(f_i) = 0.5 \times 0.8 = 0.40 \\ P'(H) + P'(T) &= 1.00 \quad \checkmark \quad (\text{normalised}) \end{aligned}$$

Under Convention B, the coin toss is now biased: heads is 50% more likely than tails during the alignment event. This is the *physically interesting* form of resonance alignment — an asymmetric modulation of probabilities that leaves the total probability normalised.

Physical interpretation

What does an $R = 1.2/0.8$ alignment event 'look like' physically? In a single coin toss, it is undetectable — the difference from fair is within statistical noise. Over a large number of independent identical events (N tosses), the expected deviation from fair grows as:

$$\begin{aligned} \text{Expected deviation from } 0.5N \text{ heads} &= N \times (0.60 - 0.50) = 0.10N \\ \text{Statistical noise (standard deviation)} &= \sqrt{N \times 0.5 \times 0.5} = 0.5\sqrt{N} \end{aligned}$$

The deviation becomes detectable above noise when $0.10N > 0.5\sqrt{N}$, i.e. $N > 25$. In principle, if resonance alignment events were sustained and asymmetric at $R = 1.2/0.8$, a sequence of 100 quantum coin tosses would show a statistically significant bias ($\sim 10\sigma$ deviation from fair). In practice, R is almost certainly much closer to 1, alignment events are transient, and the mechanism has no confirmed physical instantiation. But the toy example establishes that the formalism is *in principle testable* under controlled conditions — it is not immune to empirical challenge by construction.

[Spec] *Whether quantum events in our universe are ever actually modulated by resonance alignment is a [Spec] question. The toy example shows the formalism is self-consistent and generates calculable predictions in the right asymptotic regime. It does not constitute evidence that such modulation occurs.*

The Class IV alignment complication: $f_i = \pi$

For the Class IV universe D at $f_i = \pi$, the resonance alignment factor $R(\pi)$ presents an additional complication. Since the resonance pattern at $f_i = \pi$ is quasi-periodic (dense but non-repeating), any alignment achieved between our universe and Universe D is *transient*: the alignment phase drifts continuously without returning to its initial value. $R(\pi, t)$ is therefore time-dependent — not a constant. A full treatment would model $R(\pi, t)$ as an almost-periodic function of time, producing intermittent bursts of probability modulation rather than sustained alignment. This is consistent with the character of Class IV as structurally unstable: any resonance alignment with a Class IV universe is, by the quasi-periodicity of its resonance pattern, temporary and non-repeating.

5. Discussion

5.1 Implications for Cosmology and the Universality Assumption

The Periodic Table of Universes constitutes a direct challenge to the universality assumption — the claim that the same mathematical structures govern all of reality. By proposing that physical constants are functions of a resonance parameter that differs across universes, the framework implies that the constants we observe are the constants of a universe at resonance parameter f_0 , sitting within Class III of the resonance spectrum.

This reframing has an immediate consequence for the anthropic principle ^[5]. In the resonance framework, the fine-tuning question becomes: which range of f_i produces a universe capable of supporting complex structure? The answer is not a single point but a band within the resonance spectrum — a band characterised by a fine-structure constant in the right range for stable atomic orbitals, a gravitational constant in the right range for stellar formation, and a Planck's constant in the right range for quantum chemistry. The class boundaries derived in Section 3.5 represent

the edges of this anthropic band, grounded in the Bohr radius and Chandrasekhar mass arguments rather than in arbitrary thresholds.

5.2 Relationship to Existing Multiverse Proposals

Model	Mechanism of Universe Diversity	Relationship to Resonance Framework	Compatibility
Many-Worlds (Everett [2])	Quantum branching; universes differ in outcomes, not constants	Orthogonal: MWI addresses which outcomes occur, not what physical laws govern them. Resonance framework addresses laws, not outcomes.	Compatible — different questions
String Landscape (Susskind [3])	Vast vacuum states; each has different low-energy physics	The string landscape is a subset of Class II–III space: all landscape universes obey string-theoretic mathematics. Resonance framework is more general.	Partially subsumed
Brane-world (Randall–Sundrum [4])	Multiple 3-branes in higher-dimensional bulk; matter confined to branes; gravity leaks	Most structurally compatible: branes could be interpreted as resonance layers. Gravity leaks between branes $\approx$ resonance alignment.	Strongly compatible
Eternal Inflation (Guth [15], Linde [16])	Pocket universes nucleate from inflationary background; each may acquire different constants	Complementary: eternal inflation describes the generation mechanism; resonance framework describes the classification space.	Complementary

Table 3. Relationship between the resonance framework and four major multiverse proposals. The resonance framework is complementary to MWI and eternal inflation, partially subsumes the string landscape, and is most structurally compatible with brane-world cosmology.

5.3 The Need for New Mathematics

The resonance framework makes clear that existing mathematical tools are insufficient for a complete description of multiverse diversity. The scaling laws introduced here operate within standard real analysis — power laws are well-understood mathematical objects. But a universe at an irrational resonance parameter, whose symmetry structure may not support standard conservation laws, would require mathematical tools that go beyond anything currently available. Category theory, introduced in the companion paper as the formal language for the resonance framework, is a significant step toward this new mathematics. But even category theory, in its standard form, presupposes certain logical axioms that may not hold in Class IV universes. The full development of a multiverse mathematics — capable of representing universes with mutually incompatible internal axioms — is a long-term research programme. What the resonance

framework contributes is a clear statement of what such a mathematics must accomplish: it must encode coexisting, incompatible physical realities within a single overarching structure; it must allow meaningful comparison between them; and it must provide a language for resonance alignment that does not presuppose shared physical laws.

5.4 The Etherial Domain: Ontological Status

Three ontological positions are possible with respect to the etherial domain, and the framework is compatible with all three:

- **Position 1 — Instrumentalism:** the etherial domain is a purely formal/representational tool with no ontological commitment. It is a parameter space, like configuration space in classical mechanics or Hilbert space in quantum mechanics. On this view, asking whether the etherial domain 'really exists' is a category error.
- **Position 2 — Structural Realism:** the etherial domain is a structural ontic entity — an abstract object that exists by virtue of the relational structure it encodes, independently of whether it is observable. Spacetime, on the structural-realist reading of GR, exists as a structural entity even apart from physical fields. The etherial domain would exist in the same sense.
- **Position 3 — Mathematical Platonism:** the etherial domain is a mathematical object that exists independently of any physical universe, in the way that the number 7 or the set of real numbers exist independently of any physical instantiation.

We do not adjudicate between these positions. The framework is perfectly consistent with Position 1 (the most conservative) and does not require Position 3 (the strongest ontological claim). The mathematical utility of the etherial domain does not depend on its ontological status.

[Spec] *The choice of ontological position is a philosophical [Spec] claim, not a physical or mathematical one. The framework is agnostic.*

5.5 Resonance Alignment and Experiential Phenomena: Epistemic Boundaries

Section 4.5 introduced the resonance alignment mechanism and the toy example demonstrates its formal structure. We now state the epistemic status of the association with experiential phenomena as precisely as possible.

The formal claim is well-defined [M]: $P'(x) = P(x) \cdot R(f_i, x)$ is a mathematically coherent statement about asymmetric probability modulation, with a concrete numerical example demonstrating its internal consistency and the normalisability requirement.

The physical claim [P]: that a universe at parameter f_i achieves resonance alignment with our universe, and that quantum event probabilities are thereby modulated, requires a physical mechanism for $R(f_i, x)$ that has not been specified. This is a testable-in-principle but currently untestable hypothesis.

The phenomenological claim [Spec]: that the experiential phenomena associated with anomalous psychology (déjà vu, vivid dream states, apparitional experiences) might correspond to transient resonance alignment events is a suggestive conceptual analogy, not an empirical hypothesis. It is not testable with current methodology because: (a) there is no independent means of detecting resonance alignment; (b) the probabilistic modulation would be unmeasurably small unless R

differs dramatically from 1; (c) the phenomenological descriptions are too imprecise to serve as the basis for a physical prediction. Any reader uncomfortable with this discussion can discard Section 5.5 without affecting the integrity of the rest of the framework.

5.6 Limitations

We state the limitations of this framework explicitly and without minimisation.

- **Heuristic exponents:** the scaling exponents α , β , γ are heuristic [H]. No derivation from first principles has been provided. Different exponent choices generate different taxonomies.
- **Incomplete electromagnetic sector:** the fine-structure constant problem (Section 3.6) shows the framework as stated is incomplete. A complete framework must specify scaling laws for all ~ 26 fundamental constants simultaneously.
- **Reference circularity:** f_0 is defined as corresponding to our own universe. The framework cannot be straightforwardly extended to describe universes from a perspective external to our own physics. This is an epistemological limitation: we cannot escape our own universe's perspective.
- **No current empirical predictions:** the framework makes no empirically testable predictions with current technology. Resonance alignment modulation, even if real, would be unmeasurably small in any regime we can probe.
- **Class IV argument not a formal proof:** the irrational rotation argument is mathematically motivated but not a formal proof that conservation laws fail. The connection between irrational rotation on a circle and the failure of continuous symmetry groups in a physical universe requires a full mathematical development, of which Section 4.4 is a first sketch.

6. Future Directions

6.1 Deriving the Scaling Laws from First Principles

The most important theoretical task is to derive the scaling exponents from first principles. Three approaches are promising: dimensional analysis (requiring the scaling laws to be consistent with the fundamental dimensions of each constant); symmetry arguments (requiring scale invariance of the resonance spectrum at high f_0); and consistency with known relationships between constants — the Planck length, the Chandrasekhar mass limit, and stellar nucleosynthesis conditions. Together these constraints may uniquely determine the exponents, replacing the current [H] heuristic with a [M] derivation.

6.2 A Complete Scaling Law System

A consistent framework must specify scaling laws for all ~ 26 fundamental constants, not just three. The requirement that α_{EM} be preserved (if this is taken as a design criterion) would fix the relationship between the exponents for c , h , e , and ϵ_0 . A complete scaling law system connecting all 26 Standard Model constants is a substantial but well-defined research programme, connecting directly to the fine-tuning literature ^[17].

6.3 From Scalar to Vector Resonance Parameter

The current framework uses a scalar resonance parameter f_i . In reality, the space of physically possible universes is likely high-dimensional — parameterised by many independent quantities. A natural extension is to replace f_i with a vector $\mathbf{f}_i = (f_{i,1}, f_{i,2}, \dots, f_{i,n})$, transforming the one-dimensional resonance spectrum into a multi-dimensional resonance manifold, in which universes occupy points or regions and resonance alignment corresponds to proximity in the manifold.

6.4 Formal Axiomatisation

The framework requires a formal axiomatic foundation. The axioms should specify: (1) the topology and measure on the resonance spectrum; (2) the conditions under which the mapping function Φ is well-defined and injective; (3) the constraints on $R(\mathbf{f}_i, \mathbf{x})$; (4) the precise statement of the no-material-transfer condition. The companion paper, *Multiverse Mathematics*, proposes seven axioms for the categorical formulation; a corresponding set for the scaling law framework would complete the picture.

6.5 Categorical Extension

The companion paper, *Multiverse Mathematics*, develops the full categorical framework: universes as categories, resonance as functors, bleed-through as natural transformations, and the etherial substrate as a 2-category. Future directions include higher-dimensional functors (n -functors) for universes with different dimensionalities; monoidal structures for encoding resonance interactions as algebraic invariants; resonance geometry in which the manifold structure of the resonance spectrum is given a metric; and computational simulation using category-theoretic proof assistants.

7. Conclusion

This paper has introduced the Periodic Table of Universes: a resonance-based taxonomy of multiverse physics in which universes are classified by a dimensionless resonance parameter f_i relative to a reference value $f_0 = 1$ corresponding to our own universe. The central contribution is the mapping function $\Phi(f_i) = \{c_i, h_i, G_i, \dots\}$, which translates resonance parameter values into physical constants via power-law scaling relations, generating a structured four-class taxonomy — Classical-like, Balanced, Quantum-heavy, and Exotic.

Five specific advances have been made over the previous iteration. First, the class boundaries at $f_i \sim 10^5 \cdot f_0$ and $f_i \sim 10^{20} \cdot f_0$ are now physically motivated through the Bohr radius stability argument and the Chandrasekhar mass limit respectively, replacing the earlier purely heuristic thresholds. Second, the Class IV instability argument is made concrete with a worked example at $f_i = \pi$, showing precisely how the spectral density of the time-translation operator prevents a well-defined discrete energy eigenvalue spectrum — without which chemical structure as we know it is impossible. Third, the resonance alignment mechanism is developed with a quantum coin toss toy example with $R = 1.1$ and $R = 1.2/0.8$, making the probabilistic modulation numerically concrete, exposing the normalisability requirement, and establishing the conditions under which statistical detection would in principle be possible. Fourth, Figure 1 provides a visual schematic of the continuous resonance spectrum with all four class bands marked, oriented around our

universe's position at f_0 . Fifth, an explicit four-level epistemic map ([M], [P], [Spec], [H]) is introduced and applied throughout, ensuring the paper makes no overclaims about the status of its proposals.

The limitations of the framework are real, significant, and openly stated: the scaling exponents are heuristic [H], the electromagnetic sector is incomplete, no empirically testable predictions are currently available, and the Class IV argument is a mathematical sketch rather than a proof. These limitations define the research agenda rather than invalidating the contribution.

This paper is the first step in a two-part programme. The companion paper, *Multiverse Mathematics*, develops the formal categorical foundations: universes as categories, resonance as functors, bleed-through as natural transformations. Together, they constitute a new research programme — a mathematics of coexistence — capable of classifying and reasoning about universes with mutually incompatible physical laws, and providing a structured language for the relationships between them. The ultimate aim is not just to describe our universe but to understand its place in the full resonance spectrum of everything that may exist.

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